Eden ZERMATI

ESIEE Paris Engineering Graduate
Specialising in Data & Digital Health

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Professional Experience

Data Scientist Intern – Environmental Data (PFAS) Apr 2025 – Sep 2025
SOCOTEC Environnement – Paris, France
- Extracted, cleaned and structured environmental data from multiple sources on **PFAS** and industrial discharges.
- Conducted exploratory and statistical analyses, including comparisons between sites, correlations and spatial trends.
- Created visualisations and presented findings with **Power BI** and **QGIS** to support decision-making.
- Automated data processing using **Python**, **R** and **Excel** (VBA, macros and pivot tables).

Statistical Analysis and Meta-Analysis Intern Jun 2024 – Aug 2024
SOCOTEC Environnement – Paris, France
- Assessed feasibility and conducted a meta-analysis of epidemiological data on the association between phthalate exposure and childhood asthma.
- Collected, structured and harmonised data from the scientific literature.
- Performed statistical analyses in **R**, including pooled effect estimates, heterogeneity tests and sensitivity analyses.
- Summarised and visualised findings through tables and charts.

Data, Machine Learning & Healthcare Projects

Power BI Dashboard – Fashion Retail Sales Analysis May 2026 – Jun 2026
Personal project
- Collected and cleaned **6M+ transaction records** from a six-table dataset using **Python (pandas)**.
- Modelled the data and created **DAX** measures for net sales, return rate, average transaction value and customer loyalty segmentation.
- Designed an interactive **eight-page Power BI dashboard** covering markets, products, geography, customers, seasonality, returns, loyalty and promotions.

ECG Arrhythmia Classification – Machine Learning & Deep Learning Feb 2026 – Apr 2026
Personal project – MIT-BIH dataset
- Developed an arrhythmia detection pipeline with beat segmentation, normalisation and extraction of morphological and temporal features.
- Implemented and compared **Random Forest** and **1D CNN models in PyTorch**.
- Evaluated models on held-out patients and tuned classification thresholds. Best **ROC-AUC of 0.867**.

EMG-Controlled Articulated Prosthesis Prototype Feb 2024 – Mar 2024
Academic project – ESIEE Paris
- Acquired and preprocessed **EMG** signals and extracted physiological features.
- Applied feature engineering, signal processing and transfer learning to translate muscle signals into movement commands.

Web Development for Telemedicine Nov 2024 – Jan 2025
Academic project – ESIEE Paris
- Acquired and preprocessed physiological data from sensors.
- Structured and transferred the data to a telemedicine server.
- Performed exploratory analysis and visualised health indicators to support patient monitoring.

Education

CTI-Accredited Engineering Degree 2020 – 2025
ESIEE Paris, Université Gustave Eiffel – Noisy-le-Grand, France
Key coursework: Machine Learning & AI, data processing, data security, immunology and physiology.

International Exchange Semester – Electrical Engineering 2022
Faculty of Electrical Engineering (FEL UWB) – Plzeň, Czech Republic

Skills

Programming: Python (pandas, scikit-learn), R, SQL, VBA
Data Science: Data cleaning, EDA, PCA, statistical testing, feature engineering, machine learning
Visualisation & Tools: Microsoft Power BI, Matplotlib, Jupyter Notebook, QGIS, Dataiku, SAS
Cloud Training: AWS Knowledge: Cloud Essentials – Training Badge
Scientific Background: Cell biology, physiology, microbiology, immunology
Soft Skills: Analytical thinking, attention to detail, science communication, teamwork, empathy
Certifications: Credly profile

Languages & Interests

French: Native **English:** B2 (TOEIC 845/990) **Sports:** Fitness (5 years), modern jazz dance (13 years)